

Classful network:

Uses default subnet mask /8 /16 /24

Classless network

Uses custom network /25, 26/, 27/

Router> user execution mode

Router# privilege mode

Router(config)# global configuration mode

Router(config-if)# interface mode

Router(config-line)# line mode telnet, ssh, password

Router(config-router)# routing configuration

Change the host name

#conf t

config# hostname *mumbai* or ho *mumbai*

config# exit

Basic Commands

show startup-config

shows saved configuration

write saves

running config on startup config

copy running-config startup-config

saves running config on startup config

show clock

shows routers clock

show ip protocols

shows routing protocol details

show protocols

shows IP configuration of interface

show history

shows last 10 commands protocol

Show terminal

shows terminal setting details

show flash

shows flash settings

del flash

deletes config settings of router

erase startup-config

deletes config settings of router

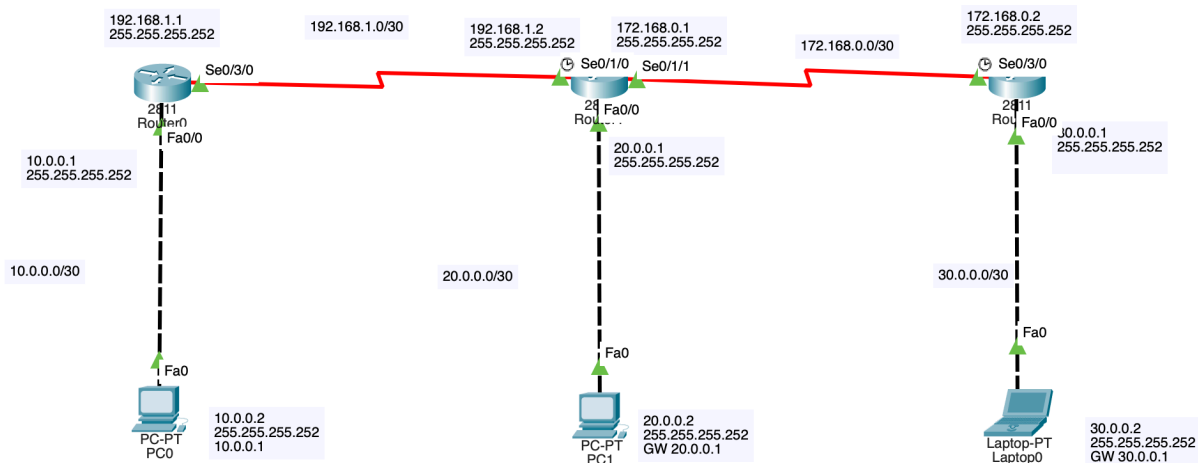
Static Routing

Notes

Static Routing

1. Next Hop

```
# conf t
config# ip route <unknown nw> <subnet mask> <next hop id>
```



```
R1(config)# ip route 20.0.0.0 255.255.255.252 192.168.1.2
R1(config)# ip route 172.168.0.0 255.255.255.252 192.168.1.2
R1(config)# ip route 30.0.0.0 255.255.255.252 192.168.1.2
```

```
R2(config)# ip route 10.0.0.0 255.255.255.252 192.168.1.1
R2(config)# ip route 30.0.0.0 255.255.255.252 172.168.0.2
```

```
R3(config)# ip route 20.0.0.0 255.255.255.252 172.168.0.1
R3(config)# ip route 192.168.1.0 255.255.255.252 172.168.0.1
R3(config)# ip route 10.0.0.0 255.255.255.252 172.168.0.1
```

2. Self Exit

```
# conf t
config# ip route <unknown nw> <subnet mask> <self serial interface >
```

```
R1(config)# ip route 20.0.0.0 255.255.255.252 se0/3/0
R1(config)# ip route 172.168.0.0 255.255.255.252 se0/3/0
R1(config)# ip route 30.0.0.0 255.255.255.252 se0/3/0
```

```
R2(config)# ip route 10.0.0.0 255.255.255.252 se0/1/0
R2(config)# ip route 30.0.0.0 255.255.255.252 se0/1/1
```

```
R3(config)# ip route 20.0.0.0 255.255.255.252 se0/3/0
R3(config)# ip route 192.168.1.0 255.255.255.252 se0/3/0
R3(config)# ip route 10.0.0.0 255.255.255.252 se0/3/0
```

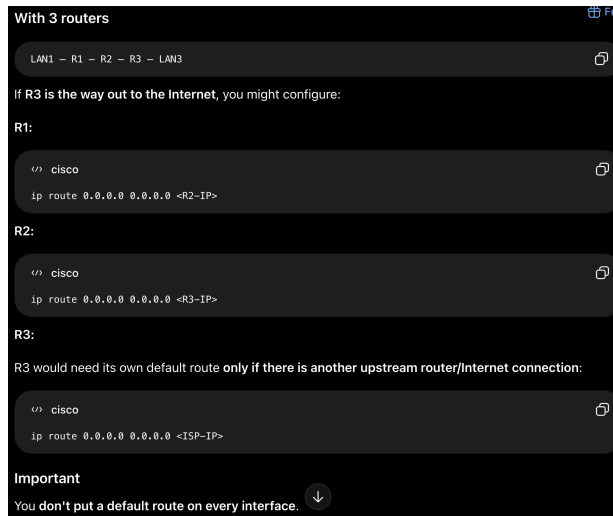
Default Routing

1.using next hop id:

```
config# ip route 0.0.0.0 0.0.0.0 <next hop id>
# do show ip route
```

2.using self exit:

```
config# ip route 0.0.0.0 0.0.0.0 s0/0
# do show ip route
```



Router Password

1. Normal Password

```
config# enable password nor@12  
config# no enable password
```

2. Secret/Encrypted

```
config# enable secret sec@123  
config# no enable secret
```

3. Console Password

- 1.To secure console mode
- 2.Must connect pc with router using console cable and to verify use terminal directly go to terminal and select default setting you will see the router>

```
config# line console 0  
config-line# password con@123  
config-line# login (must)  
config-line# no password
```

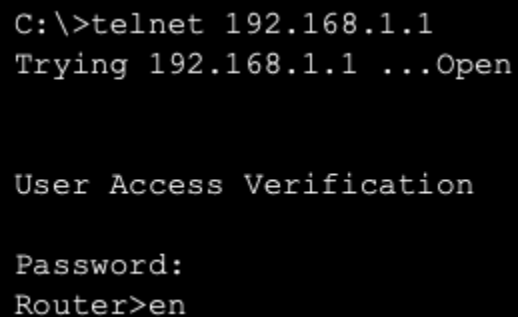
4. AUX

```
config# line aux
config-line# password aux@123
config-line# login
```

5. Telnet

1. Without setting this password you can't access the telnet

```
config# line vty 0 4
config-line# password tel@123
config-line# login
```



```
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification

Password:
Router>en
```

6. To encrypt all passwords

```
router(config)# service password-encryption
```

1. Just run this command on the start it will automatically encrypt all the passwords

SSH

Secure shell

Port no 22

Tcp

Application layer

On pc cmd

cmd# ssh -l *username connection-IP*

On router

config# username *hemant* password 123456

config# crypto key generate rsa

config# line vty 0 4

config# transport input ssh

config# login local

```

Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip domain
Router(config)#ip domain-n
Router(config)#ip domain-name corp.com
Router(config)#crypto key ge
Router(config)#crypto key generate rsa
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#ho r1
r1(config)#crypto key generate rsa
The name for the keys will be: r1.corp.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

r1(config)#line vty 0 4
*Mar 1 22:7:22.519: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 22:7:22.519: %SSH-5-ENABLED: SSH 1.5 has been enabled
r1(config-line)#transport input ssh

```

DHPC

Port no 67 and 68
UDP protocol

```

Router
config# hostname DHPC-router
config# ip dhcp pool corp
config# domain-name corp.com
config# network 192.168.1.1 255.255.255.252
config# default-router 192.168.1.1

```

```

r1>en
r1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r1(config)#int fa0/0
r1(config-if)#ip add 10.0.0.1 255.255.255.0
r1(config-if)#ip dhcp pool corp
r1(dhcp-config)#network 10.0.0.0 255.255.255.0
r1(dhcp-config)#default-router 10.0.0.1
r1(dhcp-config)#exit
r1(config)#

```

Go to IP configuration and convert all pc to static to dhcp

```

r1(config)#do show ip dhcp bi
r1(config)#do show ip dhcp binding

```

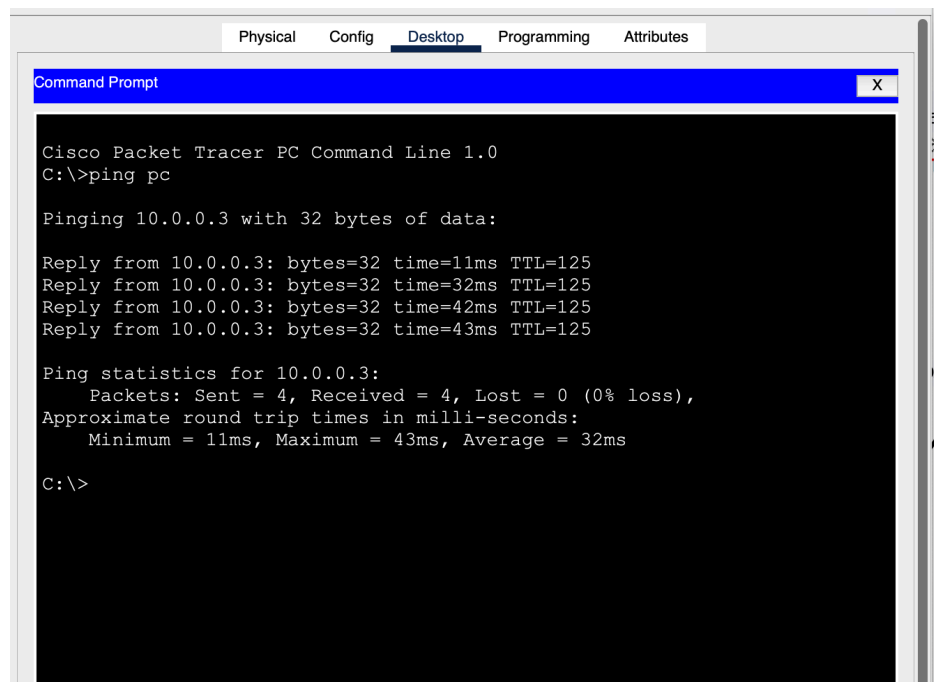
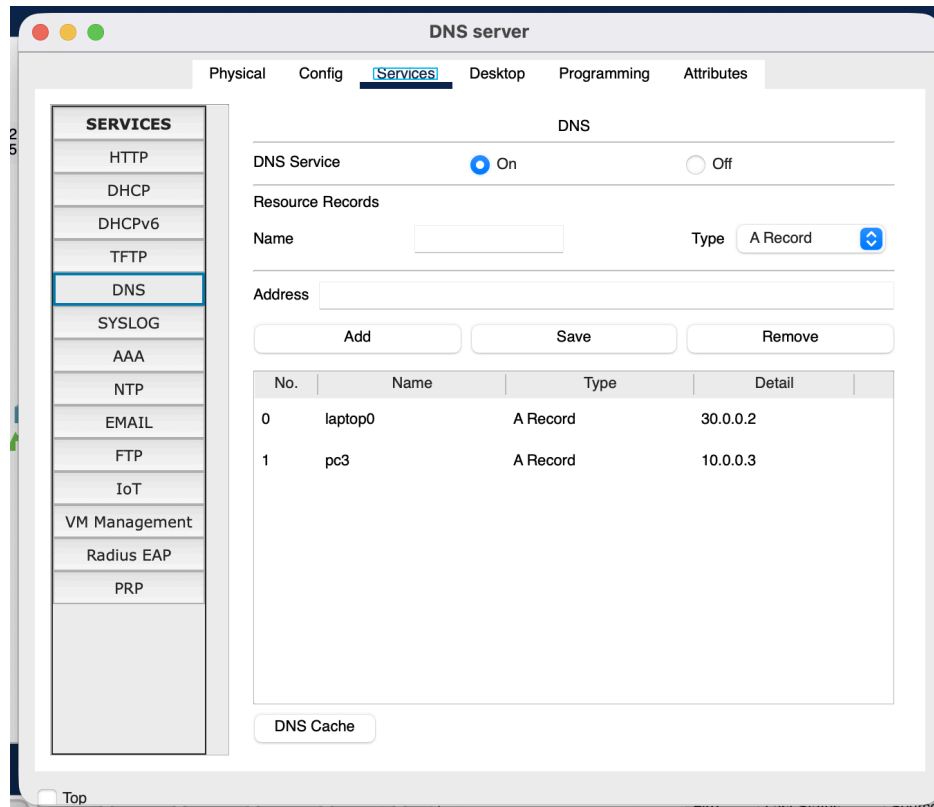
IP address	Client-ID/ Hardware address	Lease expiration	Type
10.0.0.3	0000.0CB9.BB01	--	
Automatic			
10.0.0.4	00D0.BA0D.9425	--	
Automatic			
10.0.0.5	000C.85D2.5992	--	
Automatic			

```

r1(config)#

```

DNS

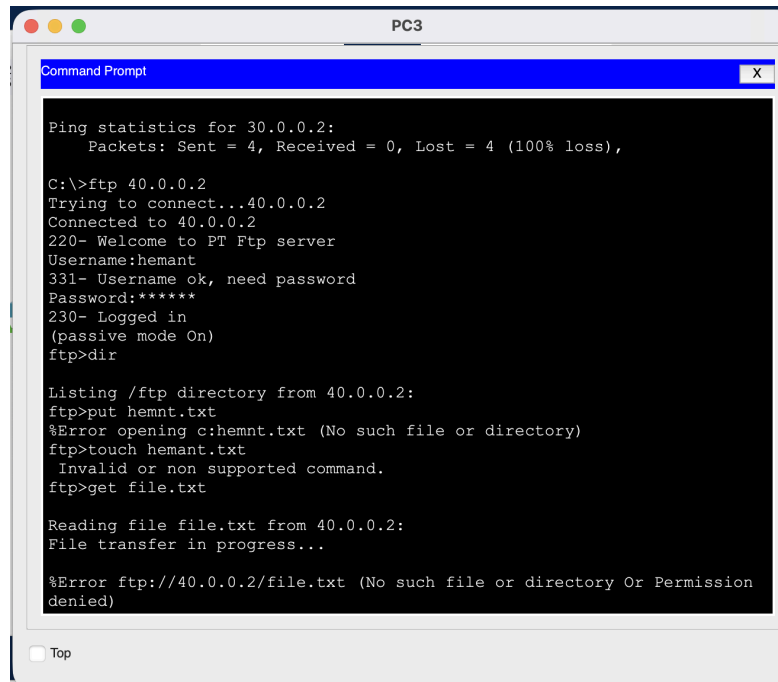


FTP

File transfer protocol

20,21 tcp

Application layer



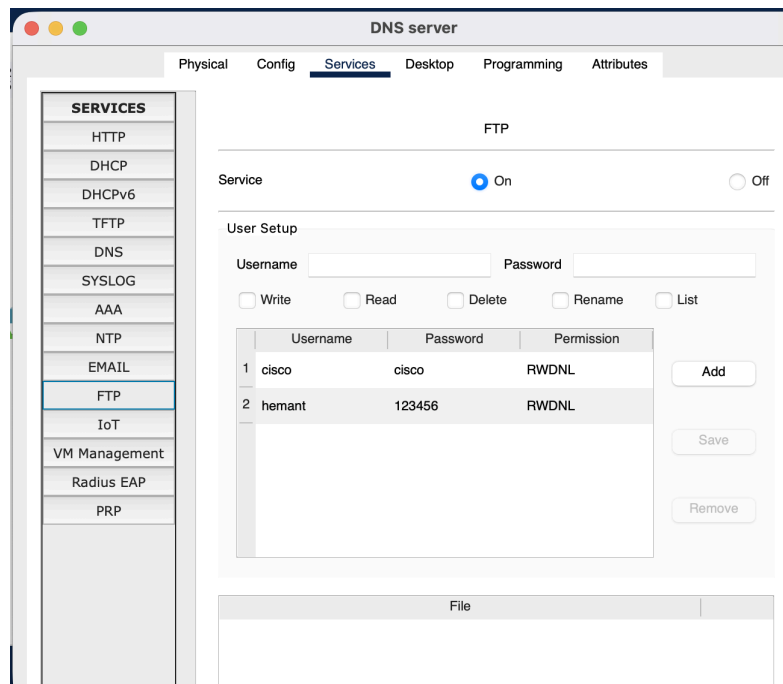
```
Command Prompt
Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ftp 40.0.0.2
Trying to connect...40.0.0.2
Connected to 40.0.0.2
220- Welcome to PT Ftp server
Username:hemant
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 40.0.0.2:
ftp>put hemnt.txt
%Error opening c:hemnt.txt (No such file or directory)
ftp>touch hemant.txt
Invalid or non supported command.
ftp>get file.txt

Reading file file.txt from 40.0.0.2:
File transfer in progress...

%Error ftp://40.0.0.2/file.txt (No such file or directory Or Permission
denied)
```



DNS server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP**
- IoT
- VM Management
- Radius EAP
- PRP

FTP

Service ☒ On ☐ Off

User Setup

Username Password

☐ Write ☐ Read ☐ Delete ☐ Rename ☐ List

	Username	Password	Permission
1	cisco	cisco	RWDNL
2	hemant	123456	RWDNL

Add Save Remove

File

POP3/SMTP

DNS server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL**
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

EMAIL

SMTP Service ☒ ON ☐ OFF

POP3 Service ☒ ON ☐ OFF

Domain Name: corp.com

User Setup

User Password

hemant
niraj

Laptop0

Physical Config **Desktop** Programming Attributes

Configure Mail

User Information

Your Name:

Email Address:

Server Information

Incoming Mail Server:

Outgoing Mail Server:

Logon Information

User Name:

Password:

PC3

Physical Config **Desktop** Programming Attributes

Mails

	From	Subject	Received
1	niraj@corp.com	happy birthday	Sat Sep 26 2026 23:57:08

Backup and Restore

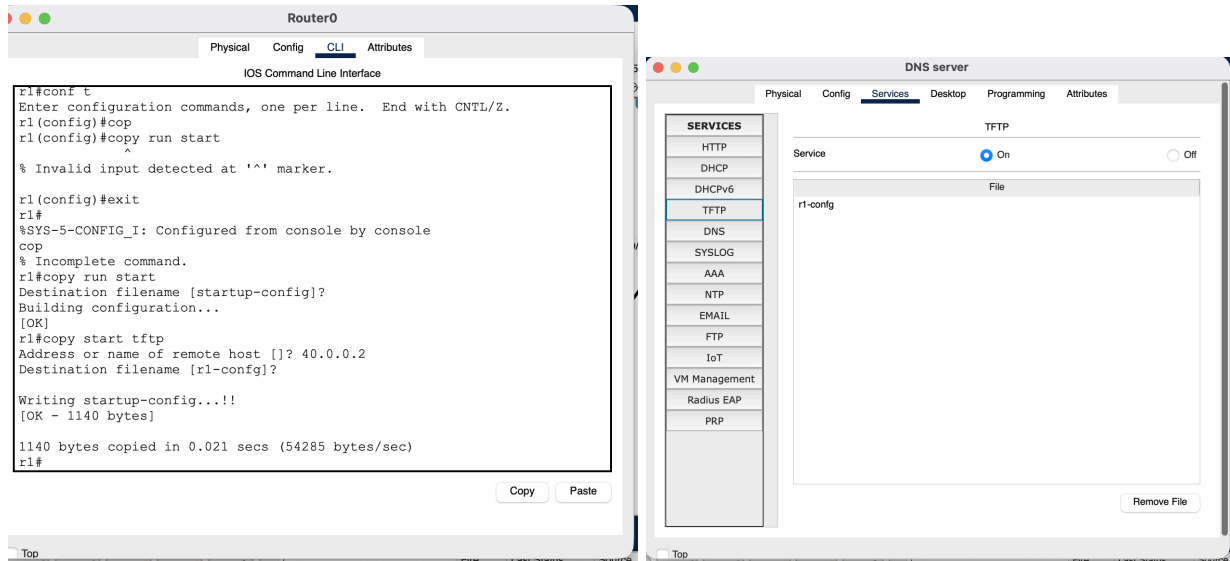
There are 2 types of backups

1. Startup-config backup & restore

It's a basic type of restore where we store the startup-config file on TFTP server and import it.

To backup

copy start tftp



To restore

copy tftp backup

```

r1#erase sta
r1#erase startup-config
Erasing the nvram filesystem will remove all configuration files!
Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
r1#show start
startup-config is not present
r1#copy tftp start
Address or name of remote host []? 40.0.0.2
Source filename []? r1-confg
Destination filename [startup-config]?

Accessing tftp://40.0.0.2/r1-confg...
Loading r1-confg from 40.0.0.2: !
[OK - 1140 bytes]

1140 bytes copied in 0.016 secs (71250 bytes/sec)
r1#show start
Using 1140 bytes

```

2. IOS backup and Restore

Here we can do the password recovery

Show flash to get the name of run-config file .bin

```
r1#show flash
```

System flash directory:

File	Length	Name/status
------	--------	-------------

3	33591768	2800nm-advipservicesk9-mz.151-4.M4.bin
---	----------	--

2	28282	sigdef-category.xml
---	-------	---------------------

1	227537	sigdef-default.xml
---	--------	--------------------

[33847587 bytes used, 221896413 available, 255744000 total]

249856K bytes of processor board System flash (Read/Write)

Copy flash on tftp server

```
r1#copy flash tftp
```

Source filename []? 40.0.0.2

Address or name of remote host []?

```
r1#copy flash tftp
```

Source filename []? 2800nm-advipservicesk9-mz.151-4.M4.bin

Address or name of remote host []? 40.0.0.2

Destination filename [2800nm-advipservicesk9-mz.151-4.M4.bin]?

Writing 2800nm-advipservicesk9-

mz.151-4.M4.bin...!!

!!

!!

!!

!!

!!

!!

!!

!!

!!

!!

[OK - 33591768 bytes]

33591768 bytes copied in 8.232 secs (428449 bytes/sec)

```
r1#
```

Now to **delete flash** use **del flash** and **reload** to restore

```
r1#del flash
Delete filename []?2800nm-advipservicesk9-mz.151-4.M4.bin
Delete flash:/2800nm-advipservicesk9-mz.151-4.M4.bin? [confirm]

r1#reload
Proceed with reload? [confirm]
System Bootstrap, Version 12.4(1r) [hqluong 1r], RELEASE SOFTWARE (fc1)
Copyright (c) 2005 by cisco Systems, Inc.

Initializing memory for ECC
..
C2800 processor with 524288 Kbytes of main memory
Main memory is configured to 64 bit mode with ECC enabled

Readonly ROMMON initialized

Boot process failed...

The system is unable to boot automatically. The BOOT
environment variable needs to be set to a bootable
image.
rommon 1 > |
```

Use command **tftpdnld** to setup tftp server

After that fill this all variables

The following variables are REQUIRED to be set for tftpdnld:

- IP_ADDRESS: The IP address for this unit
- IP_SUBNET_MASK: The subnet mask for this unit
- DEFAULT_GATEWAY: The default gateway for this unit
- TFTP_SERVER: The IP address of the server to fetch from
- TFTP_FILE: The filename to fetch

```
rommon 2 > IP_ADDRESS=192.168.1.1
rommon 3 > IP_SUBNET_MASK=255.255.255.0
rommon 4 > DEFAULT_GATEWAY=192.168.1.1
rommon 5 > TFTP_SERVER=40.0.0.2
rommon 6 > TFTP_FILE=
rommon 6 > TFTP_FILE=2800nm-advipservicesk9-mz.151-4.M4.bin
rommon 7 > tftpdnld

      IP_ADDRESS: 192.168.1.1
      IP_SUBNET_MASK: 255.255.255.0
      DEFAULT_GATEWAY: 192.168.1.1
      TFTP_SERVER: 40.0.0.2
      TFTP_FILE: 2800nm-advipservicesk9-mz.151-4.M4.bin
Invoke this command for disaster recovery only.
WARNING: all existing data in all partitions on flash will be lost!

Do you wish to continue? y/n: [n]: |
```

Do you wish to continue? y/n: [n]: y

.....[TIMED OUT]

TFTP: Operation terminated.

rommon 16 > TFTP_TIMEOUT=720000

rommon 17 > tftpdnld

IP_ADDRESS: 10.0.0.1

IP_SUBNET_MASK: 255.255.255.0

DEFAULT_GATEWAY: 10.0.0.1

TFTP_SERVER: 10.0.0.5

TFTP_FILE: 2800nm-advipservicesk9-mz.151-4.M4.bin

Invoke this command for disaster recovery only.

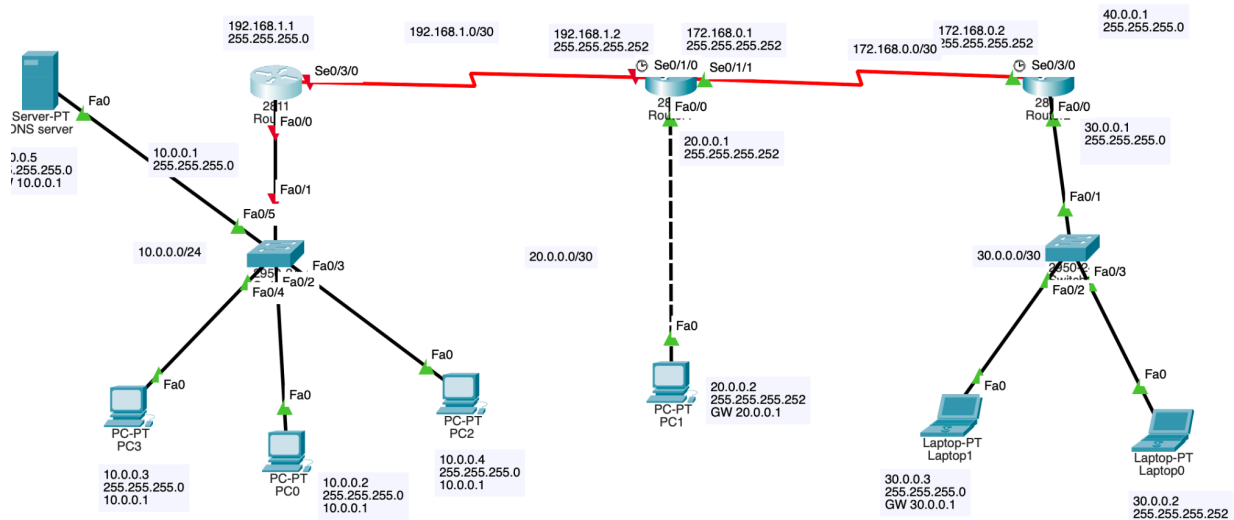
WARNING: all existing data in all partitions on flash will be lost!

Do you wish to continue? y/n: [n]: y

.....[TIMED OUT]

TFTP: Operation terminated.

rommon 18 >



3.Password Recovery

Set password and then copy the run file to the start we need start file

```
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#enable secret 123456
r3(config)#exit
r3#
```

```
r3#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
```

To enter in **rommon** mode turn off and on and quickly click ctrl + c in terminal

0x2142 is the recovery mode

Then go to recovery mood and then boot the file

```
rommon 1 > confreg 0x2142
rommon 2 > boot
System Bootstrap, Version 12.4(1r) [hqluong 1r], RELEASE SOFTWARE (fc1)
Copyright (c) 2005 by cisco Systems, Inc.
```

Copy the start file to run file and use no enable password to remove password and use **Confreg 0x2102 or config-register 0x2142** and then **reload** the it

```
Router>en
Router#copy start run
Destination filename [running-config]?

969 bytes copied in 0.416 secs (2329 bytes/sec)
r3#
%SYS-5-CONFIG_I: Configured from console by console
```

```
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#no enable secret
r3(config)#config-re
r3(config)#config-register 0x2102
r3(config)#do wr
Building configuration...
[OK]
r3(config)#do reload
```

Dynamic Routing

RIP

Router 1

Gateway of last resort is not set

```
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      10.0.0.0/24 is directly connected, FastEthernet0/0
L      10.0.0.1/32 is directly connected, FastEthernet0/0
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/30 is directly connected, Serial0/3/1
L      192.168.1.1/32 is directly connected, Serial0/3/1
```

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#router rip

Router(config-router)#network 10.0.0.0

Router(config-router)#network 192.168.0.1

Router(config-router)#version

% Incomplete command.

Router(config-router)#version 2

Router(config-router)#

Router 2

Gateway of last resort is not set

```
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      20.0.0.0/30 is directly connected, FastEthernet0/0
L      20.0.0.1/32 is directly connected, FastEthernet0/0
      172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.168.0.0/30 is directly connected, Serial0/1/1
L      172.168.0.1/32 is directly connected, Serial0/1/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.1.0/30 is directly connected, Serial0/1/0
L      192.168.1.2/32 is directly connected, Serial0/1/0
```

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#router rip

Router(config-router)#network 20.0.0.0

Router(config-router)#network 172.168.0.0

Router(config-router)#network 192.168.1.0

Router(config-router)#version 2

Router(config-router)#

Router 3

```

    10.0.0.0/24 is subnetted, 1 subnets
S      10.0.0.0/24 [1/0] via 172.168.0.1
    20.0.0.0/30 is subnetted, 1 subnets
S      20.0.0.0/30 [1/0] via 172.168.0.1
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/24 is directly connected, FastEthernet0/0
L      30.0.0.1/32 is directly connected, FastEthernet0/0
    172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.168.0.0/30 is directly connected, Serial0/3/0
L      172.168.0.2/32 is directly connected, Serial0/3/0
    192.168.1.0/30 is subnetted, 1 subnets
S      192.168.1.0/30 [1/0] via 172.168.0.1

r3#
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#router rip
r3(config-router)#network 30.0.0.0
r3(config-router)#network 172.168.0.0
r3(config-router)#version 2
r3(config-router)#
```

To remove RIP just use **no router rip** at every router

And to verify if router setup is done correctly use

do show ip router rip

EIGRP

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/24 is directly connected, FastEthernet0/0
L    10.0.0.1/32 is directly connected, FastEthernet0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/30 is directly connected, Serial0/3/1
L    192.168.1.1/32 is directly connected, Serial0/3/1
```

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#no au
Router(config-router)#no auto-summary
Router(config-router)#
```

```
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/30 is directly connected, FastEthernet0/0
L    20.0.0.1/32 is directly connected, FastEthernet0/0
172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.168.0.0/30 is directly connected, Serial0/1/1
L    172.168.0.1/32 is directly connected, Serial0/1/1
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/30 is directly connected, Serial0/1/0
L    192.168.1.2/32 is directly connected, Serial0/1/0
```

```
Router#router eigrp 1
      ^
% Invalid input detected at '^' marker.
```

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 1
Router(config-router)#network 20.0.0.0
Router(config-router)#network 172.168.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 192.168.1.1 (Serial0/1/0) is up:
new adjacency
```

```
10.0.0.0/24 is subnetted, 1 subnets
S   10.0.0.0/24 [1/0] via 172.168.0.1
20.0.0.0/30 is subnetted, 1 subnets
S   20.0.0.0/30 [1/0] via 172.168.0.1
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   30.0.0.0/24 is directly connected, FastEthernet0/0
L   30.0.0.1/32 is directly connected, FastEthernet0/0
172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C   172.168.0.0/30 is directly connected, Serial0/3/0
L   172.168.0.2/32 is directly connected, Serial0/3/0
192.168.1.0/30 is subnetted, 1 subnets
S   192.168.1.0/30 [1/0] via 172.168.0.1

r3(config)#router eigrp 1
r3(config-router)#network 30.0.0.0
r3(config-router)#network 172.168.0.0
r3(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 172.168.0.1 (Serial0/3/0) is up:
new adjacency

r3(config-router)#no au
r3(config-router)#no auto-summary
```

Loopback Interface

Loopback Interface: is a logical interface

```
conf t
int lo0
ip add 5.0.0.1 255.0.0.0
```

```
exit
int lo1
ip add 9.0.0.1 255.0.0.0
```

```
exit
config#router eigrp 10
        network 5.0.0.0
        network 9.0.0.0
```

```
do sh ip route
```

```
goto pc5
```

```
ping 9.0.0.1
ping 5.0.0.1
```

OSPF

```
Router(config)#router ospf 1
Router(config-router)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter ar
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EG
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-I
inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
        10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.0.0.0/24 is directly connected, FastEthernet0/0
L        10.0.0.1/32 is directly connected, FastEthernet0/0
        192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, Serial0/3/1
L        192.168.1.1/32 is directly connected, Serial0/3/1
```

```
Router(config-router)#network 10.0.0.0 0.0.0.255
% Incomplete command.
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
```

Gateway of last resort is not set

```
        20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/30 is directly connected, FastEthernet0/0
L        20.0.0.1/32 is directly connected, FastEthernet0/0
        172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.168.0.0/30 is directly connected, Serial0/1/1
L        172.168.0.1/32 is directly connected, Serial0/1/1
        192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, Serial0/1/0
L        192.168.1.2/32 is directly connected, Serial0/1/0
```

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 20.0.0.0 0.0.0.3 area 0
Router(config-router)#network 172.168.0.0 0.0.0.3 area 0
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#
00:48:05: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/1/0
LOADING to FULL, Loading Done

00:49:02: %OSPF-5-ADJCHG: Process 1, Nbr 172.168.0.2 on Serial0/1/1
LOADING to FULL, Loading Done
```

```
10.0.0.0/24 is subnetted, 1 subnets
S    10.0.0.0/24 [1/0] via 172.168.0.1
20.0.0.0/30 is subnetted, 1 subnets
S    20.0.0.0/30 [1/0] via 172.168.0.1
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/24 is directly connected, FastEthernet0/0
L    30.0.0.1/32 is directly connected, FastEthernet0/0
172.168.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.168.0.0/30 is directly connected, Serial0/3/0
L    172.168.0.2/32 is directly connected, Serial0/3/0
192.168.1.0/30 is subnetted, 1 subnets
S    192.168.1.0/30 [1/0] via 172.168.0.1

r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#router ospf 1
r3(config-router)#network 172.168.0.0 0.0.0.3 area 0
r3(config-router)#network
00:48:56: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.2 on Serial0/3/0
LOADING to FULL, Loading Done

% Incomplete command.
r3(config-router)#network 30.0.0.0 0.0.0.3 area 0
```

Redistribution

RIP-EIGRP

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no au
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#
```

```
middle>en
middle#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
middle(config)#router rip
middle(config-router)#version 2
middle(config-router)#network 192.168.1.0
middle(config-router)#red
middle(config-router)#redistribute ei
middle(config-router)#redistribute eigrp ?
    <1-65535>  Autonomous system number
middle(config-router)#redistribute eigrp 1 ?
    metric  Metric for redistributed routes
    <cr>
middle(config-router)#redistribute eigrp 1 metric ?
    <0-16>      Default metric
    transparent Transparently redistribute metric
middle(config-router)#redistribute eigrp 1 metric 1 ?
    <cr>
middle(config-router)#redistribute eigrp 1 metric 1
middle(config-router)#exit
```

```
middle(config)#router eigrp 1
middle(config-router)#no a
middle(config-router)#no auto-summary
middle(config-router)#network 172.168.0.0
middle(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 172.168.0.2 (Serial0/1/1)
new adjacency
```

```
middle(config-router)#red
middle(config-router)#redistribute rip met
middle(config-router)#redistribute rip metric 1 1 1 1 1
middle(config-router)#
```

```
r3>en
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#router ospf 1
r3(config-router)#network 172.168.0.0 0.0.0.3 area 0
r3(config-router)#network 30.0.0.0 0.0.0.255 area 0
r3(config-router)#exit
r3(config)#no router ospf 1
r3(config)#router eigrp 1
r3(config-router)#network 172.168.0.0
r3(config-router)#network 30.0.0.0
r3(config-router)#no aut
r3(config-router)#no auto-summary
r3(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 172.168.0.1 (Serial0/3/0)
new adjacency
```

RIP-OSPF

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no au
Router(config-router)#no auto-summary
Router(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.1.0
Router(config-router)#
```

```
middle>en
middle#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
middle(config)#router rip
middle(config-router)#version 2
middle(config-router)#network 192.168.1.0
middle(config-router)#red
middle(config-router)#redistribute ospf 1 met
middle(config-router)#redistribute ospf 1 metric 3
middle(config-router)#exit
middle(config)#router ospf 1
middle(config-router)#network 172.168.0.0 0.0.0.3 me
middle(config-router)#network 172.168.0.0 0.0.0.3 area 0
middle(config-router)#redi
middle(config-router)#redistribute rip ?
    metric          Metric for redistributed routes
    metric-type     OSPF/IS-IS exterior metric type for redistributed routes
    subnets        Consider subnets for redistribution into OSPF
    tag             Set tag for routes redistributed into OSPF
    <cr>
middle(config-router)#redistribute rip subnets
middle(config-router)#
```

```
r3>en
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#router ospf 1
r3(config-router)#network 172.168.0.0 0.0.0.3 area 0
r3(config-router)#network 30.0.0.0 0.0.0.255 area 0
r3(config-router)#
```

EIGRP-OSPF

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no router rip
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.0.0.255 area 0
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#
04:44:15: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.2 on Serial0/3
LOADING to FULL, Loading Done
|

middle(config)#router ospf 1
middle(config-router)#network 192.168.1.0 0.0.0.3 area 0
middle(config-router)#red
middle(config-router)#redistribute eigrp metric ?
% Unrecognized command
middle(config-router)#redistribute eigrp 1 metric ?
<0-16777214> OSPF default metric
middle(config-router)#redistribute eigrp 1 metric 500 ?
metric-type OSPF/IS-IS exterior metric type for redistributed
subnets    Consider subnets for redistribution into OSPF
tag          Set tag for routes redistributed into OSPF
<cr>
middle(config-router)#redistribute eigrp 1 metric 500 subnets ?
tag Set tag for routes redistributed into OSPF
<cr>
middle(config-router)#redistribute eigrp 1 metric 500 subnets
middle(config-router)#exit
middle(config)#router eigrp
% Incomplete command.
middle(config)#router eigrp 1
middle(config-router)#network 172.168.0.0
middle(config-router)#
```

```

middle(config-router)#red
middle(config-router)#redistribute ospf 1 ?
  match   Redistribution of OSPF routes
  metric  Metric for redistributed routes
  <cr>
middle(config-router)#redistribute ospf 1 metric ?
  <1-4294967295>  Bandwidth metric in Kbits per second
middle(config-router)#redistribute ospf 1 metric 500 ?
  <0-4294967295>  EIGRP delay metric, in 10 microsecond units
middle(config-router)#redistribute ospf 1 metric 500 333 ?
  <0-255>  EIGRP reliability metric where 255 is 100% reliable
middle(config-router)#redistribute ospf 1 metric 500 333 255 ?
  <1-255>  EIGRP Effective bandwidth metric (Loading) where 255 i
loaded
middle(config-router)#redistribute ospf 1 metric 500 333 255 255
  <1-65535>  EIGRP MTU of the path
middle(config-router)#redistribute ospf 1 metric 500 333 255 255
  match   Redistribution of OSPF routes
  <cr>
middle(config-router)#redistribute ospf 1 metric 500 333 255 255
middle(config-router)#no auto
middle(config-router)#no auto-summary

```

```

r3>en
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#router ospf 1
r3(config-router)#network 172.168.0.0 0.0.0.3 area 0
r3(config-router)#network 30.0.0.0 0.0.0.255 area 0
r3(config-router)#exit
r3(config)#no router ospf 1
r3(config)#router eigrp 1
r3(config-router)#network 172.168.0.0
r3(config-router)#network 30.0.0.0
r3(config-router)#no aut
r3(config-router)#no auto-summary

```

Numbered Standard ACL

Access Control Lists

- Syntax for a Numbered Standard ACL:

```
access-list <ID#> <action> <source>
```

- Syntax for a Numbered Extended ACL:

```
access-list <ID#> <action> <protocol> <source> <destination>
```

Access Control Lists

- Syntax for a Numbered Standard ACL:

```
access-list <ID#> <action> <source>
```

- Syntax for a Numbered Extended ACL:

```
access-list <ID#> <action> <protocol> <source> <destination>
```

- <source> or <destination>

- Either are defined with: <IP Addresses> [Port]

- To match a Subnet: <NetID> <Wildcard Mask>
- To match a single IP address: host <#.#.#.#>
- To match all IP addresses: any

Extended ACL

**Named
Standard ACL**

Extended ACL

